

RISC-V VeeR EL2 Based Real-Time Data Buffering SoC

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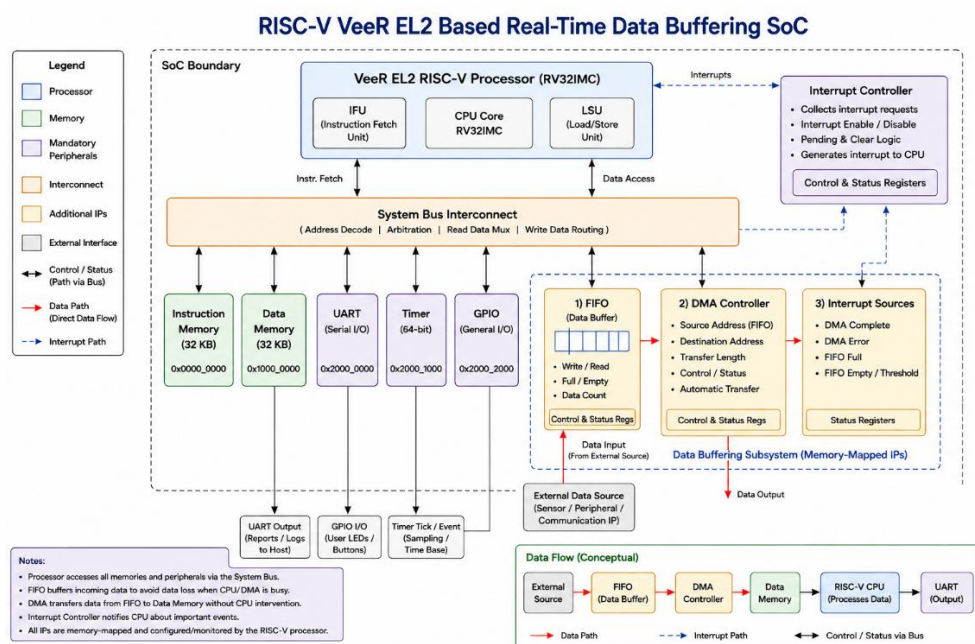
1. System Architecture

The proposed SoC is organized around the VeeR EL2 RISC-V processor as the central processing unit, connected to instruction memory, data memory, mandatory peripherals, and additional IP blocks through a shared system bus. The mandatory peripherals include UART, Timer, and GPIO, while the three additional IPs are FIFO, DMA Controller, and Interrupt Controller. The FIFO temporarily buffers incoming data, the DMA Controller transfers buffered data to Data Memory with minimal processor involvement, and the Interrupt Controller manages events such as DMA completion, transfer errors, and FIFO status. The processor accesses the memories and IPs through memory-mapped registers and controls the overall data-transfer operation.

Data is supplied by an external data source or testbench, stored temporarily in the FIFO, and transferred to Data Memory through the DMA Controller. The RISC-V processor configures the data-transfer subsystem, processes the transferred data, and reports transfer status through UART.

2. Block Diagram

The block diagram shows the VeeR EL2 RISC-V processor connected to memories, mandatory peripherals, and the three additional IPs through the System Bus Interconnect. The real-time data path is:



Data Source → FIFO → DMA Controller → Data Memory

The Interrupt Controller monitors DMA and FIFO events and generates an interrupt to the RISC-V processor when a programmed event occurs. The processor reads the transfer status and processed data through the system bus and reports the results through UART.

3. Block Diagram Description

3.1 VeeR EL2 RISC-V Processor

The VeeR EL2 processor is the central processing unit of the SoC. It executes the application, configures the FIFO and DMA Controller, handles interrupts, and processes the transferred data.

3.2 System Bus Interconnect

The System Bus connects the processor, memories, mandatory peripherals, and additional IPs. It performs address decoding and routes processor read/write transactions to the selected memory or peripheral.

3.3 Instruction Memory

Instruction Memory stores the application program executed by the VeeR EL2 processor.

3.4 Data Memory

Data Memory stores application data, variables, and data blocks transferred by the DMA Controller.

3.5 UART

UART provides serial output for displaying DMA transfer status, received data information, and system messages.

3.6 Timer

The Timer provides timing functionality and periodic timing events required by the application.

3.7 GPIO

GPIO provides general-purpose digital input/output and can be used for basic control or transfer-status indication.

3.8 FIFO

The FIFO temporarily buffers incoming data before it is transferred to memory. It provides status information such as **empty**, **full**, and **data count** conditions to manage data availability and overflow.

3.9 DMA Controller

The DMA Controller transfers data from the FIFO to Data Memory without requiring the processor to handle each data word individually. It uses configurable source address, destination address, and transfer-length registers to perform the programmed transfer.

3.10 Interrupt Controller

The Interrupt Controller collects interrupt requests from the DMA Controller and FIFO. It manages interrupt enable, pending, and clear status and generates an interrupt to the RISC-V processor for events such as DMA completion, DMA error, or FIFO threshold/full conditions.

3.11 External Data Source

The External Data Source represents a sensor, peripheral, communication interface, or testbench-generated data stream used to verify the buffering and transfer operation.

3.12 Data Transfer Flow

The data flow is:

Data Source → FIFO → DMA Controller → Data Memory

The processor configures the FIFO and DMA through the system bus, while the Interrupt Controller notifies the processor when important transfer events occur. The processor then reads and processes the transferred data and reports the result through UART.

4. Memory Mapping

The SoC uses a **memory-mapped architecture** in which the RISC-V processor accesses memories and peripherals using dedicated address ranges. The System Bus decodes the processor address and selects the corresponding memory or peripheral. Each block is assigned a separate address region to avoid conflicts and simplify peripheral integration.

A preliminary memory map, following the same aligned-window approach as the reference document, is:

Sl. No.	Block	Base Address (Hex)	End Address (Hex)	Size	Description
1	Instruction Memory	0x0000_0000	0x0000_7FFF	32 KB	Stores program instructions
2	Data Memory	0x0001_0000	0x0001_7FFF	32 KB	Stores application data and DMA-transferred data
3	UART	0x1000_0000	0x1000_0FFF	4 KB	Serial communication and result reporting
4	Timer	0x1000_1000	0x1000_1FFF	4 KB	Timing and periodic events
5	GPIO	0x1000_2000	0x1000_2FFF	4 KB	General-purpose digital I/O
6	FIFO	0x1000_3000	0x1000_3FFF	4 KB	Buffers incoming data

7	DMA Controller	0x1000_4000	0x1000_4FFF	4 KB	Transfers data from FIFO to Data Memory
8	Interrupt Controller	0x1000_5000	0x1000_5FFF	4 KB	Manages DMA and FIFO interrupt requests

The FIFO, DMA Controller, and Interrupt Controller are accessed through their respective memory-mapped register interfaces. As in the reference architecture, these addresses should be treated as a preliminary map and finalized after checking the actual VeeR EL2 RTL configuration and resolving any processor-local memory-region conflicts.

5. Interface Specifications

The SoC uses a memory-mapped interface in which the VeeR EL2 processor accesses memories and peripherals through the System Bus Interconnect. The bus performs address decoding and routes read/write transactions to the selected block, following the reference architecture

Interface	Connected Blocks	Main Signals / Information	Function
Processor–Bus Interface	VeeR EL2 ↔ System Bus	Address, Write Data, Read Data, Read/Write, Valid, Ready/Response	Transfers processor requests to memories and peripherals
Instruction Memory Interface	Bus ↔ Instruction Memory	Address, Read Enable, Read Data	Fetches program instructions
Data Memory Interface	Bus ↔ Data Memory	Address, Read/Write, Write Data, Read Data	Stores application and DMA-transferred data
UART Interface	Bus ↔ UART	Address, Control, TX/RX Data, Status	Reports system and transfer results
Timer Interface	Bus ↔ Timer	Configuration, Counter, Status	Provides timing and periodic events
GPIO Interface	Bus ↔ GPIO	GPIO Input, GPIO Output, Control	General-purpose digital I/O
FIFO Interface	External Source ↔ FIFO	Data In, Write Enable, Read Enable, Data Out, Full, Empty, Count	Buffers incoming data
FIFO–DMA Interface	FIFO ↔ DMA Controller	FIFO Data, Read Enable, Empty, Count	Supplies buffered data to DMA
DMA–Memory Interface	DMA ↔ Data Memory	Address, Read/Write, Data, Enable, Transfer Count	Transfers data from FIFO to memory
Interrupt Interface	FIFO/DMA ↔ Interrupt Controller ↔ RISC-V	IRQ, Enable, Pending, Clear, Acknowledge	Signals DMA completion, errors, and FIFO events
Control/Status Interface	RISC-V ↔ Additional IPs	Memory-mapped Registers	Configures and monitors FIFO, DMA, and interrupt controller

Important Data and Control Paths

Data Path:

External Data Source → FIFO → DMA Controller → Data Memory

Control Path:

RISC-V → System Bus → FIFO / DMA / Interrupt Controller

Interrupt Path:

FIFO / DMA Events → Interrupt Controller → RISC-V

The reference architecture similarly separates the processor's bus-based control/status access from the application-specific hardware data path.

6. Bus Protocol Details

The SoC uses a memory-mapped system bus to provide communication between the VeeR EL2 RISC-V processor, memories, mandatory peripherals, and additional IP blocks. The bus interconnect decodes the processor address and selects the corresponding slave device. The VeeR EL2 processor acts as the bus master for instruction and data transactions, while Instruction Memory, Data Memory, UART, Timer, GPIO, FIFO, DMA Controller, and Interrupt Controller operate as bus slaves.

6.1 Bus Architecture

- The VeeR EL2 processor acts as the main bus master for instruction and data transactions.
- The System Bus Interconnect performs address decoding and generates the appropriate slave-select signal.
- Instruction Memory, Data Memory, UART, Timer, GPIO, FIFO, DMA Controller, and Interrupt Controller are connected as bus slaves.
- Read transactions return data from the selected slave to the processor.
- Write transactions transfer processor-generated data to the selected slave.
- The interconnect provides the required response and transaction-completion signals.

6.2 Read Transaction

A typical read transaction follows these steps:

1. The processor provides the target address and read control information.
2. The bus decodes the address and selects the corresponding slave.
3. The selected slave provides the requested data.
4. The bus routes the read data back to the processor.
5. The transaction completes when the bus indicates a valid response.

6.3 Write Transaction

A typical write transaction follows these steps:

1. The processor provides the target address, write control information, and write data.
2. The bus decodes the address and selects the corresponding slave.
3. The selected slave accepts the write data.

4. The addressed register or memory location is updated.
5. The transaction completes with the appropriate bus response.

6.4 Peripheral Access

All additional IPs are accessed through memory-mapped registers. The RISC-V processor can access FIFO control/status registers, configure DMA registers, and read or clear Interrupt Controller status through the system bus.

The main control path is:

RISC-V → System Bus → FIFO / DMA Controller / Interrupt Controller

The actual data-transfer path is:

External Data Source → FIFO → DMA Controller → Data Memory

6.5 Bus Error Handling

An invalid or unmapped processor address should generate an appropriate bus error response rather than selecting a valid peripheral. This prevents unintended access to memories or peripheral registers.

6.6 Protocol Configuration

The final bus protocol, signal widths, timing, and handshake details will be finalized after confirming the bus configuration of the faculty-provided VeeR EL2 RTL. The reference architecture identifies AHB-Lite as the intended shared-bus approach, using address decoding, read-data multiplexing, and HREADY-based transaction completion.

7. Architecture Summary

The proposed RISC-V VeeR EL2 Based Real-Time Data Buffering SoC integrates the VeeR EL2 RISC-V processor with Instruction Memory, Data Memory, UART, Timer, GPIO, and three additional IP blocks: FIFO, DMA Controller, and Interrupt Controller. The processor controls the system through a memory-mapped AHB-Lite system bus, while the FIFO buffers incoming data and the DMA Controller transfers the buffered data to Data Memory with minimal processor involvement. The Interrupt Controller monitors important events such as DMA completion, transfer errors, and FIFO conditions and notifies the processor through interrupt signals.

The primary data path is External Data Source → FIFO → DMA Controller → Data Memory, while the RISC-V processor configures the peripherals, processes the transferred data, and reports the results through UART. The complete architecture will be verified through RTL simulation and embedded software execution using normal data transfers, multiple data blocks, FIFO full conditions, DMA completion, DMA error conditions, and interrupt handling.